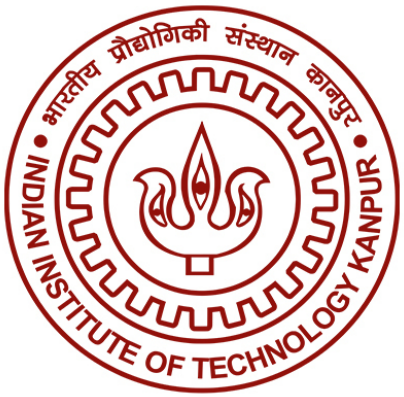


SPA617: Radio Astronomy Lab

RF components

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1. Aim

To understand the operation ans experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in a RF receiver chain.

2. Objectives

The objectives of the experiment are:

1. To characterize an RF filter, identity its type, and determine its cuttoff frequencies or pass-band.
2. To demostrate a quarter-wave open-circuit stub as a notch filter.
3. To observe frequency conversion in a RF mixer and estimate its conversion loss.
4. To measure the power division the insertion loss of a power splitter/combiner.
5. To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.
6. To estimate the input reflection coefficient and return loss of the filter.

3. Theory

3.1 RF Components in an RF Receiver Chain

An RF receiver chain consists of several components that perform different functions on the received radio-frequency (RF) signal. The major components studied in this experiment are the RF filter, quarter-wave open-circuit stub, mixer, power splitter/combiner, and RF amplifier.

A filter selects the required frequency range and suppresses unwanted frequency components. A mixer translates a signal from one frequency to another, generally producing an intermediate-frequency (IF) signal. An RF amplifier increases the power of a weak signal, while a power splitter or combiner distributes a signal into multiple paths or combines signals from different paths.

The performance of these components can be characterized using parameters such as insertion loss, reflection coefficient, return loss, conversion loss, power division, gain, harmonic generation, the 1-dB compression point, and the third-order intercept point (TOI).

3.2 RF Filter

An RF filter is a frequency-selective device that allows a desired range of frequencies to pass while attenuating unwanted frequency components. Depending on its frequency response, an RF filter can be classified as a low-pass, high-pass, band-pass, or band-reject (notch) filter.

3.2.1 Low-Pass Filter

A low-pass filter allows frequencies below its cutoff frequency f_c to pass while attenuating frequencies above f_c.

3.2.2 High-Pass Filter

A high-pass filter allows frequencies above its cutoff frequency f_c to pass while attenuating frequencies below f_c.

3.2.3 Band-Pass Filter

A band-pass filter allows frequencies between a lower cutoff frequency f_1 and an upper cutoff frequency f_2 to pass, while attenuating frequencies outside this range. The centre frequency is denoted by f_0.

3.2.4 Band-Reject or Notch Filter

A band-reject filter attenuates frequencies within a specified stopband while allowing frequencies outside the stopband to pass. A narrow band-reject filter is commonly referred to as a notch filter. The notch frequency is denoted by f_0.

3.2.5 Insertion Loss

The transmission performance of a filter can be characterized using its insertion loss. Let P_{in} be the power measured without the filter and P_{out} be the power measured after inserting the filter. The insertion loss is given by

$$IL = P_{\text{in}} - P_{\text{out}} \quad \text{dB}. \tag{1}$$

A smaller value of insertion loss indicates better transmission through the filter. The cutoff frequencies are generally determined from the -3 dB points relative to the minimum

insertion loss in the passband.

3.3 Filter Reflection Coefficient and Return Loss

When the input impedance Z_{in} of a filter is different from the characteristic impedance Z_0 of the RF system, a portion of the incident signal is reflected from the filter input.

The voltage reflection coefficient is defined as

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \tag{2}$$

where V_{inc} and V_{ref} are the incident and reflected voltage amplitudes, respectively.

The magnitude $|\Gamma|$ represents the fraction of the incident voltage amplitude that is reflected. For a perfectly matched system,

$$Z_{\text{in}} = Z_0 \quad \Rightarrow \quad |\Gamma| = 0. \tag{3}$$

For complete reflection,

$$|\Gamma| = 1. \tag{4}$$

The return loss is given by

$$RL = -20 \log_{10} |\Gamma| \quad \text{dB}. \tag{5}$$

A well-matched filter has a small reflection coefficient and a large return loss, whereas a poorly matched filter has a larger reflection coefficient and a smaller return loss.

During the experimental measurement, the maximum and minimum voltage amplitudes at the Tee junction can be used to estimate the incident and reflected voltages:

$$V_{\text{inc}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}, \tag{6}$$

$$V_{\text{ref}} = \frac{V_{\text{max}} - V_{\text{min}}}{2}. \tag{7}$$

Therefore, the magnitude of the reflection coefficient is

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\text{max}} - V_{\text{min}}}{V_{\text{max}} + V_{\text{min}}}. \tag{8}$$

3.4 Quarter-Wave Open-Circuit Stub

An open-ended coaxial stub connected in shunt through a Tee connector can behave as a resonator and produce a notch in the transmission response.

At the quarter-wave resonant frequency, the open circuit at the far end of the stub is transformed approximately into a low impedance at the Tee junction. Consequently, the RF signal is strongly attenuated at this frequency, producing a notch in the transmission response.

The physical length of the stub is approximately one-quarter of the guided wavelength:

$$L = \frac{\lambda}{4}. \tag{9}$$

Taking into account the velocity factor VF of the coaxial cable,

$$L = \frac{VF\,c}{4f_0}, \tag{10}$$

where L is the physical length of the stub, VF is the velocity factor of the cable, c is the speed of light in vacuum, and f_0 is the expected notch frequency.

Hence, the expected notch frequency is

$$f_0 = \frac{VF\,c}{4L}. \tag{11}$$

The measured notch frequency may differ slightly from the calculated value because of connector length and open-end effects.

3.5 RF Mixer

An RF mixer is a three-port nonlinear device consisting of an RF port, a local-oscillator (LO) port, and an intermediate-frequency (IF) port.

When signals having frequencies f_{RF} and f_{LO} are applied to the RF and LO ports, respectively, the nonlinear operation of the mixer produces sum and difference frequency components.

The principal mixing products are

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} + f_{\mathrm{LO}}, \tag{12}$$

and

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} - f_{\mathrm{LO}}, \tag{13}$$

where the second expression assumes $f_{\mathrm{RF}} > f_{\mathrm{LO}}$.

In a receiver, the required IF component is selected using a filter.

The performance of a mixer can be characterized using its conversion loss. If P_{RF} is the RF input power and P_{IF} is the measured power of the selected IF component, then

$$\text{Conversion Loss} = P_{\mathrm{RF}} - P_{\mathrm{IF}} \quad \text{dB}. \tag{14}$$

3.6 Power Splitter and Combiner

A power splitter divides an input RF signal into two or more output paths. For an ideal two-way equal-power splitter, the input power is divided equally between the two output ports. Consequently, each output is 3 dB below the input power.

For a practical splitter, a small additional insertion loss is also present.

If P_{in} is the input power and P_{out1} and P_{out2} are the powers measured at the two output ports, the losses are

$$\mathrm{Loss}_1 = P_{\mathrm{in}} - P_{\mathrm{out1}}, \tag{15}$$

and

$$\mathrm{Loss}_2 = P_{\mathrm{in}} - P_{\mathrm{out2}}. \tag{16}$$

A power combiner performs the reverse operation by combining signals from two or more input paths into a common output. The resulting output depends on the frequencies, amplitudes, and relative phases of the input signals.

3.7 RF Amplifier

An RF amplifier increases the power of an RF signal. If P_{in} and P_{out} are the input and output powers of the amplifier, respectively, the small-signal gain is

$$G = P_{\mathrm{out}} - P_{\mathrm{in}} \quad \text{dB}. \tag{17}$$

The gain of an amplifier can be measured as a function of frequency to determine its

operating frequency range and frequency response.

A practical RF amplifier is nonlinear at sufficiently high signal levels. Its output can be represented by the nonlinear relation

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots . \tag{18}$$

For a sinusoidal input

$$x(t) = A\cos(\omega t), \tag{19}$$

the nonlinear terms generate additional frequency components. These include the fundamental frequency, the second harmonic at 2ω, the third harmonic at 3ω, and higher-order harmonics.

Thus, nonlinear operation of an RF amplifier results in harmonic distortion and intermodulation products.

3.8 1-dB Compression Point

At low input power, an amplifier operates in its small-signal region and its gain remains approximately constant. As the input power is increased, the amplifier eventually approaches its nonlinear or saturation region, causing its gain to decrease.

The **1-dB compression point** is defined as the input power at which the amplifier gain has decreased by 1 dB from its small-signal gain. The corresponding output power is denoted by P_{out}.

The 1-dB compression point therefore provides a measure of the input power level beyond which the amplifier no longer maintains its small-signal gain.

3.9 Third-Order Intercept Point

When two closely spaced input tones at frequencies f_1 and f_2 are applied to a nonlinear amplifier, the output contains the fundamental components, harmonics, and intermodulation products.

The third-order intermodulation products closest to the fundamental frequencies occur at

$$f_{\text{IM3,1}} = 2f_1 - f_2, \tag{20}$$

and

$$f_{\mathrm{IM3,2}} = 2f_2 - f_1. \tag{21}$$

The fundamental output power increases approximately with a slope of $1:1$ with increasing input power, whereas the third-order intermodulation output increases approximately with a slope of $3:1$.

If these two trends are extrapolated, their intersection defines the **third-order intercept point (TOI)**.

The TOI is a theoretical measure of the third-order linearity of an amplifier. In practice, the amplifier normally reaches compression before the theoretical TOI point is reached.

3.10 RF Tee Connector and $50\,\Omega$ Termination

An RF Tee connector provides three RF connections and is used to connect the signal path to the device under test and the measurement system. In this experiment, it is particularly important for the filter reflection measurement and the quarter-wave open-circuit stub experiment.

A $50\,\Omega$ termination provides a matched load for the RF system and minimizes unwanted reflections. Unused ports of a splitter or combiner should therefore be terminated with a $50\,\Omega$ load.

The input impedance of the spectrum analyser is also $50\,\Omega$. Hence, the maximum permissible input power of the spectrum analyser must not be exceeded.

3.11 RF Signal Generator

The RF signal generator provides controlled RF signals of specified frequency and power to the device under test. It is used as the input source for transmission measurements of the filter, quarter-wave stub, power splitter, and amplifier.

For mixer characterization, two signal generators provide the RF and LO signals. For the two-tone amplifier experiment, the outputs of two signal generators are combined using an RF power combiner before being applied to the amplifier.

3.12 Spectrum Analyser

A spectrum analyser measures the frequency spectrum and power of RF signals. It is used to observe the transmitted power from the device under test and to identify different

frequency components.

In the filter experiment, the spectrum analyser measures P_{out} at different frequencies. In the mixer experiment, it is used to identify the sum and difference frequency components. For amplifier characterization, it is used to measure the fundamental output, harmonics, and third-order intermodulation products.

The spectrum analyser has a $50\,\Omega$ input impedance, and its maximum permissible input power must not be exceeded.

3.13 Digital Storage Oscilloscope

The digital storage oscilloscope (DSO) is used to measure the voltage variation required for estimating the reflection coefficient of the filter.

In the Tee-connector method, the oscilloscope measures the maximum and minimum voltage amplitudes produced by the interference of incident and reflected waves. These measurements are used to calculate

$$V_{\text{inc}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}, \tag{22}$$

and

$$V_{\text{ref}} = \frac{V_{\text{max}} - V_{\text{min}}}{2}. \tag{23}$$

The magnitude of the reflection coefficient is then obtained from

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\text{max}} - V_{\text{min}}}{V_{\text{max}} + V_{\text{min}}}. \tag{24}$$

For this measurement, the oscilloscope input impedance is set to $1\,\text{M}\Omega$ so that the oscilloscope does not significantly load the Tee junction.

3.14 RF Cables, Adapters and Attenuators

RF cables and adapters provide the required connections between the signal generator, device under test, and measuring instruments. Since cable and connector losses can affect RF power measurements, the same cables and adapters should be used for reference and device-under-test measurements wherever possible.

Attenuators may be used when required to reduce the RF power reaching the measurement instrument and to ensure that the maximum safe input power of the spectrum

analyser is not exceeded.

Before making any connection, the operating-frequency range and maximum safe input power of every RF component should be verified. “‘

4. Equipment

The following equipment was used during the experiment

- Radio-frequency signal generator
- Spectrum analyzer
- Oscilloscope / DSO
- Filter
- Mixer
- Power splitter/Combiner
- RF amplifier and the required DC power supply
- RF Tee connector
- OPen-ended coaxial stub
- Short reference coaxial cable
- 50 Ω terminators
- Suitable RF cables, adapters, and attenuators

5. Observation

5.1 Filter characterization

The measured values of input power without the filter and the output power with the filter along with corresponding insertion loss are given in table 1.

Table 1: Insertion loss Vs Frequency

Input Frequency	$\mathbf{P_{in}}$ (dBm) without filter	$\mathbf{P_{out}}$ with filter(dBm)	$\|\mathbf{\Gamma}\|$
50	-4.25	-4.10	0.15
100	-4.40	-4.13	0.37
150	-4.50	-4.40	0.10
200	-4.70	-4.20	0.50
250	-4.90	-4.40	0.50
300	-4.95	-4.42	0.53
350	-5.02	-4.50	0.12
400	-5.60	-4.60	1.00
450	-5.74	-4.73	0.99
500	-6.31	-4.95	1.36
550	-8.04	-4.65	3.39
600	-10.25	-4.95	5.30
650	-23.80	-4.72	19.08
700	-34.50	-4.67	29.83
750	-45.40	-4.70	40.70

The insertion loss remains small at lower frequencies, indicating that the filter allows low-frequency signals to pass with minimal attenuation. As the frequency increases, the insertion loss begins to rise significantly, indicating the transition from the passband to the stopband. The cutoff frequency is approximately 550 MHz, where the insertion loss is close to 3 dB. Beyond the cutoff frequency, the insertion loss increases rapidly, reaching approximately 40.73 dB at 750 MHz. Thus, the measured response exhibits the expected low-pass filter characteristics, with good transmission in the passband and strong attenuation in the stopband.

5.2 Quarter-Wave Stub

The measured transmission response of the open-ended coaxial stub is given in Table 2. We observe that the attenuation increases rapidly to 15.60 dBm as the frequency approaches to 58 MHz.

Hence, the experimentally determined notch frequency of the stub is approximately **58 MHz**, with an attenuation of about **15.60 dB**at this frequency. Beyond the notch frequency, the attenuation decreases, confirming the frequency-selective nature of the quarter-wave stub.

Table 2: response of the quarter-wave opencircuit stub.

Frequency(MHz)	P_{in} (dBm)	P_{out} (dBm)	Attenuation (dBm)
50	0.00	-2.70	2.70
52	0.00	-2.42	2.42
54	0.00	-3.16	3.16
56	0.00	-9.53	9.53
58	0.00	-15.60	15.60
60	0.00	-6.30	6.30
62	0.00	-1.20	1.20
64	0.00	-1.80	1.80
66	0.00	-3.85	3.85

5.3 Filter reflection and return loss

The measured voltage values and calculated reflection parameters are given in Table 3. At 50 MHz, the measured reflection coefficient is around 9.5 % corresponding to a return

Table 3: filter reflection coefficient and return loss

Frequency range(MHz)	V_{max} (V)	V_{min} (V)	V_{inc} (V)	V_{ref} (V)	RL(dB)
30	10	0.97	1.07	0.10	-20.63

loss of approximately RL = -20.63 dB. This indicates relatively good impedance matching at the measured frequency.

5.4 Mixer characterization

The mixer characterization shows that the conversion loss varies with the operating frequency. The conversion loss decreases from 7.20 dBm to 5.50 dBm as the IF output power increases from 10.20 dBm to 8.50 dBm, indicating improved conversion efficiency at the latter operating points. The nearly constant conversion loss of 5.50 dBm for the last two measurements suggests stable mixer performance over that frequency range.

Table 4: Mixer characterization

$f_{RF}(MHz)$	$f_{LO}(MHz)$	Expected f_{IF} (MHz)	Measured f_{IF} (MHz)	P_{IF} (dBm)	Conversion Loss
30	10	20	19.99	-10.02	7.02
40	10	30	29.97	-8.50	5.50
50	10	40	39.99	-8.50	5.50

5.5 Power splitter characteristic

At 60 MHz, the power splitter divides the input signal equally between its two output ports, with both outputs measuring 4.63 dBm for an input power of 0 dBm. The measured insertion loss of 4.63 dB is higher than the theoretical 3 dB loss of an ideal 2-way splitter, indicating additional practical losses due to the splitter and measurement setup. The equal output powers demonstrate good power balance between the two output ports

Table 5: Power splitter characterization

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	$Loss_1$ (dBm)	$Loss_1$ (dBm)
60	0.00	-4.63	-4.63	4.63	4.63

5.6 Power combiner characterization

At 60 MHz, two equal input signals of 3 dBm were applied to the power combiner, resulting in an output power of 7.75 dBm. The measured combining loss was approximately 4.75 dB. The difference between the measured and ideal performance can be attributed to the practical insertion loss of the combiner, impedance mismatch, connector and cable losses, and measurement uncertainties.

Table 6: response of the quarter-wave opencircuit stub.

Frequency(MHz)	P_1 (dBm)	P_2 (dBm)	P_{out} (dBm)
60	-3.00	-3.00	-7.75

5.7 Amplifier characterization

The amplifier gain shows a gradual decrease with increasing frequency, from 24.15 dB at 100 MHz to 22.19 dB at 600 MHz.

Table 7: Amplifier characterization.

Input Frequency (MHz)	Input Power (dBm)	Output Power (dBm)	Gain
100	-20	4.15	24.15
150	-20	3.80	23.80
200	-20	3.70	23.70
250	-20	3.42	23.42
300	-20	3.42	23.42
350	-20	3.14	23.14
400	-20	2.90	22.90
450	-20	2.91	22.91
500	-20	2.32	22.32
550	-20	2.50	22.50
600	-20	2.19	22.19

5.8 Hramonics and 1-dB compression point

Setting the 100 MHz frequency within the amplifier operating band we measured the respective output power gain respectively with the change of input power. At 17.80 dBm power level the output power drops by 1.23 dBm (≈ 1 dBm).

Table 8: Hramonics and 1-dB compression point.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-20	4.03	24.03
-19	5.02	24.03
-18	6.00	24.00
-17	7.00	24.00
-16	8.00	24.00
-15	9.00	24.00
-14	9.97	23.97
-13	10.97	23.97
-12	11.95	23.95
-11	12.93	23.93

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-10	13.90	23.90
-9	14.86	23.86
-8	15.77	23.77
-7	16.56	23.56
-6	17.23	23.23
-5	17.80	22.80
-4	18.22	22.22
-3	18.52	21.52
-2	18.70	20.70

5.9 Third-order intercept (TOI)

The output power at fundamental and the third order product has been mapped. The fundamental shows a slope approximately 1:1, while IM3 output followed a 3:1. The extrapolated intersection of them known as Third order intercept (TOI) has been calculated and turned out to be 28.12 dBm for the fundamental and 8.43 dBm for IM3.

Table 9: Third order intercept.

P_{in} per tone (dBm)	P_{out} fundamental (dBm)	$P_{3rd-order}$
-10	10.89	-25.90
-9	11.64	-22.55
-8	12.54	-19.05
-7	13.47	-15.70
-6	14.30	-13.30
-5	15.10	-10.30
-4	15.87	-7.90
-3	16.60	-5.70
-2	17.22	-3.80
-1	17.80	-2.30
0	18.25	-1.34
1	18.67	-0.08

6. Calculation and Plots

6.1 Low pass filter response plot

The cutoff frequency is approximately **550 MHz**, where the insertion loss is close to 3 dB. Beyond the cutoff frequency, the insertion loss increases rapidly, reaching approximately **40.73 dB** at **750 MHz**. Thus, the measured response exhibits the expected low-pass filter characteristics, with good transmission in the passband and strong attenuation in the stopband.

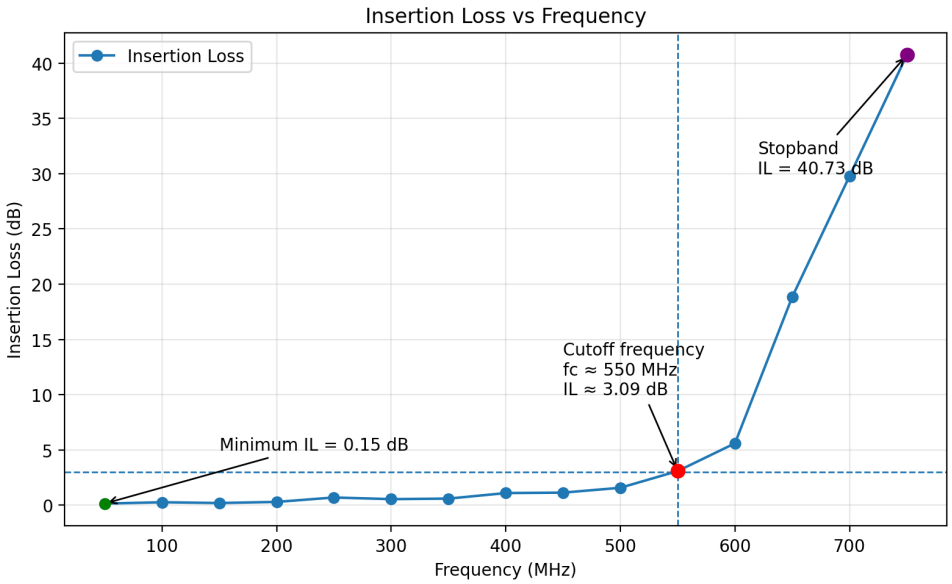


Figure 1: Insertion loss response of the Low pass filter

6.2 Quarter wave stub response

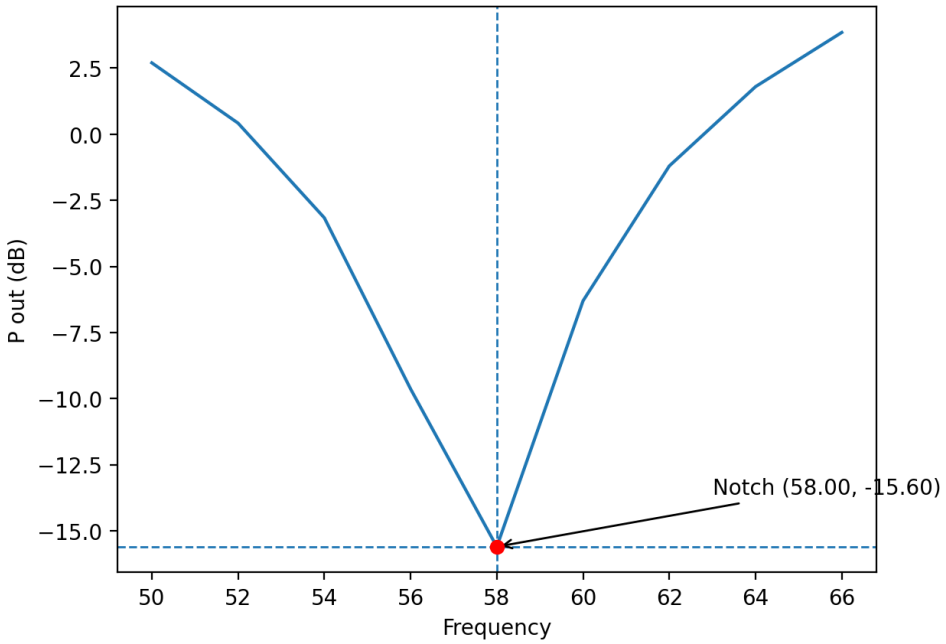


Figure 2: Notch response of the Quarter wave stub

The experimentally determined notch frequency of the stub is approximately **58 MHz**, with an attenuation of about **15.60 dB**at this frequency. Beyond the notch frequency, the attenuation decreases, confirming the frequency-selective nature of the quarter-wave stub.

The expected notch frequency value is given by

$$f_0 = \frac{VF \times c}{4L}$$

Substituiting the values of VF = 0.7153 and the measured length of cable 92.5 cm we get

$$\boxed{f_0 = 57.99\ MHz}$$

Comparing it to the observed notch frequency where the minimum a output power

$$\%Error = \frac{|f_{0,theory} - f_{0,obs}|}{f_{0,theory}}$$

$$\boxed{\%Error = 0.002}$$

6.3 Amplifier characterisation

The amplifier gain shows a gradual decrease with increasing frequency, from 24.15 dB at 100 MHz to 22.19 dB at 600 MHz. The small fluctuations in gain are likely due to experimental errors and measurement uncertainties.

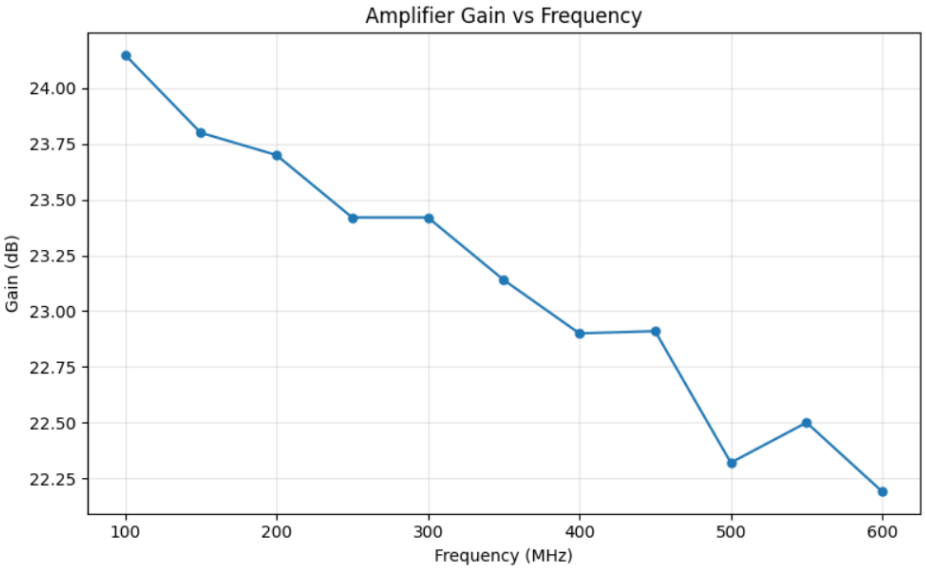


Figure 3: Amplifier gain corresponding to different frequencies

6.4 Harmonics and 1-dB compression point

The amplifier input vs output power has been plotted and we obtained the 1-dB compression point corresponds to the P_{out} = 17.50 dBm where the gain dropped by exactly 1 dBm from the plot.

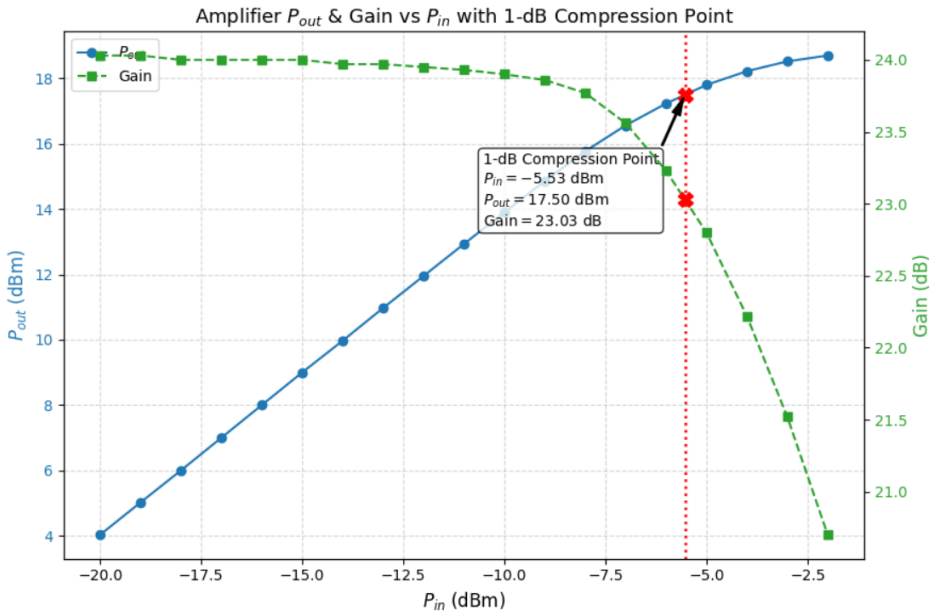


Figure 4: Amplifier gain with 1-dB insertion point

6.5 Third-order intercept

The input power vs output power at fundamental and the third order product has been plotted. The fundamental shows a slope approximately 1:1, while IM3 output followed a 3:1. The extrapolated intersection of that is known as Third order intercept (TOI) has been recorded from the plot. 28.12 dBm for the fundamental and 8.43 dBm for IM3.

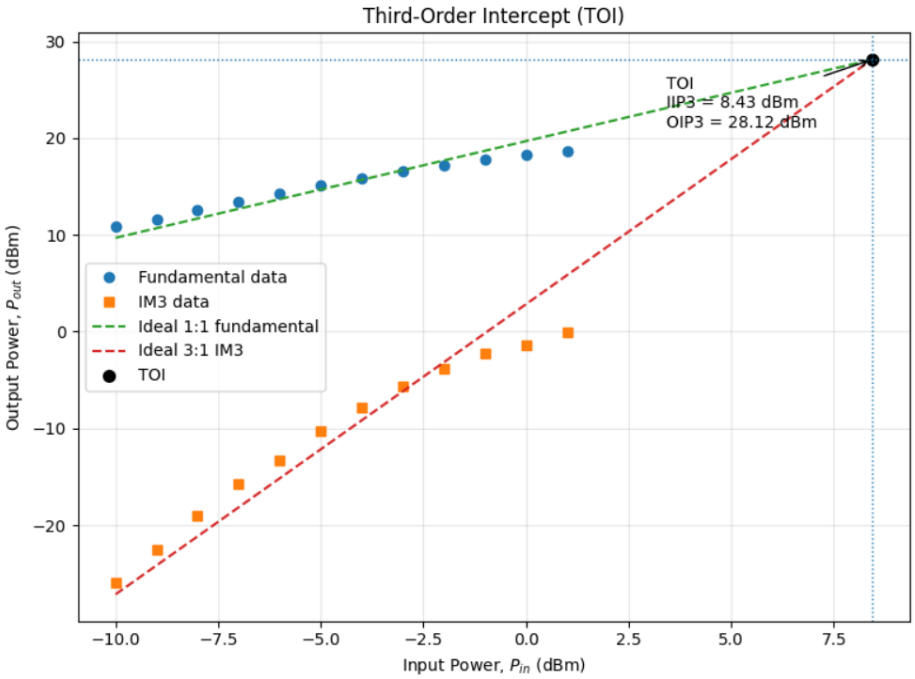


Figure 5

7. Results and Discussion

- **Filter Characterization** The measured frequency response confirms the expected low-pass characteristics of the filter. The signal experiences relatively low attenuation at lower frequencies, while the attenuation increases significantly beyond the cutoff region. The measured cutoff frequency was approximately $f_c = 550\,\text{MHz}$ with insertion loss around $IL = 3.01dBm$, and the frequency corresponding to maximam loss 20 dB was around $f_s = 750\,\text{MHz}$. These results demonstrate the filter's frequency-selective behaviour, allowing lower-frequency signals to pass while effectively attenuating frequencies above approximately 546 MHz.

- **Quarter wave stub** In our observation the notch frequency was obtained at 58 MHz with minimum power output of -15.60 dBm. The theoretically expected value of notch frequency was calculated to be 57.99 MHz which matches almost perfectly with observed frequency with a minimum error of 0.002 %.

- **Mixer characterization** The observations confirms proper frequency conversion from the RF and LO signals to the IF signal. The measured conversion loss was found to be approximately 5.5–7.2dB, indicating reasonable practical mixer performance with some insertion and conversion losses.

- **Power splitter characterization** The power splitter successfully divided the input signal equally between the two output ports, with both outputs measuring 4.63 dBm at 60 MHz for a 0 dBm input. The measured loss of 4.63 dB per port is higher than the ideal 3.01 dB splitting loss, indicating additional practical losses.

- **Power combiner characterization** The power combiner characterization at 60 MHz showed that two input signals of 3 dBm each were combined to produce an output power of 7.75 dBm. The measured combining loss was approximately 4.75 dBm. The deviation from ideal combining performance can be attributed to the practical insertion loss of the combiner, impedance mismatch, connector and cable losses, and measurement uncertainties. Overall, the results demonstrate the expected signal-combining behaviour of the practical power combiner.

- **Amplifier characterization** The amplifier was characterized by measuring the output power and corresponding gain at different input frequencies for a fixed input power. The measured gain was plotted as a function of frequency to study the frequency response of the amplifier. In addition, the 1-dB compression point (P1dB) was determined ($P_{out} = 17.50\ dBm$ where the output drops exactly bt 1 dBm) by observing the deviation of the actual gain from its ideal linear gain by 1 dB. Finally, third-order intercept (TOI) was obtained ($IIP3 = 8.43\ dBm\ OIP3 =$

28.12 dBm) by applying two input tones and measuring the fundamental and third-order intermodulation (IM3) products. The fundamental and IM3 output powers were plotted against input power with ideal 1:1 and 3:1 slopes, respectively, and their extrapolated intersection was used to determine the IIP3 and OIP3.